Maths With Math Mentor @ 078 399 0193

For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

MATHEMATICS **4004/2**

PAPER 2 2 hours 30 minutes

NOVEMBER 2025 SESSION

Additional materials:
Mathematical Instruments
Mathematical Tables
Non programmable Electronic Calculator
Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.
Answer **all questions** in Section A and **any four questions** from Section B.
If you use more than one booklet, fasten them together.
All working must be clearly shown on the same sheet as the rest of the answer.
Omission of essential working will result in loss of marks.
If the degree of accuracy is not specified in the question and if the answer is not exact, the answer should be given correct to three significant figures. Answers in degrees should be given correct to one decimal place.
Mathematical tables and Non programmable electronic calculators may be used to evaluate explicit numerical expressions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 9 printed pages.

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[Turn over

Section A [52 marks]

Answer **all** questions in this section

1 **(a)** The Universal Set, ξ has subsets P and Q such that:

$\xi = \{1; 2; 3; 4; 5; ...; 10\}$

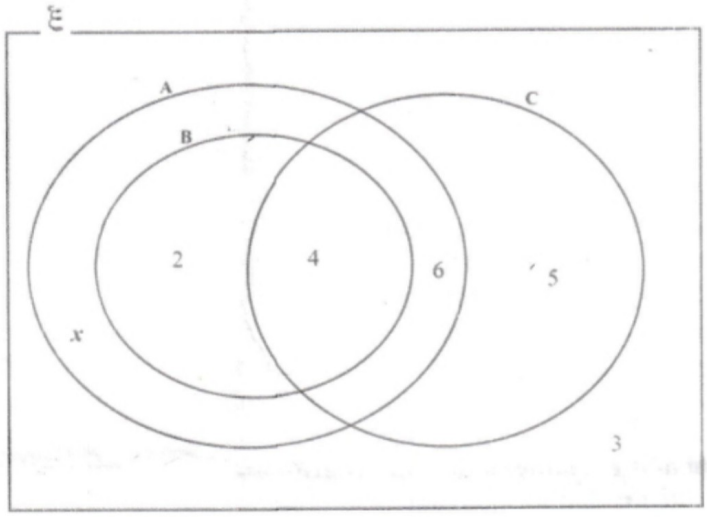
$P = \{2; 3; 5; 7\}$

Q is a set of factors of 42.

(i) List all elements of $P \cap Q$. [2]

(ii) Find $n(P \cup Q)'$. [2]

(b)



The Venn diagram shows the Universal Set ξ and three subsets A, B and C.

The number of elements in each region of the sets is shown.

Given that $n(A) = 22$,

find,

(i) the value of x, [2]

(ii) $n(A \cap C)$. [2]

2 **(a)** Expand $(3 - 2y)^2$. [2]

(b) Factorise $2x^2 - 3x + 1$. [2]

(c) Solve the equation $3(2x - 1) - (2 - x) = 7$. [3]

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(d) Solve the simultaneous equations:

$$3m + n = -5$$
$$m = 1 - 3n$$

[3]

3 **(a)** A vendor bought 120 cabbages at \$ 7 200 for sale. The vendor sold all the cabbages at \$ 45 each because the market was flooded.

Calculate the percentage loss made by the vendor. [3]

(b) The exchange rate between the Zimbabwean dollar and the United States of American dollar on a certain day was ZWL\$ 85 TO USA\$1.

Express USA \$170 in Zimbabwean dollars on that day. [2]

(c) Calculate the Simple Interest if \$ 6000 is invested at 7,5 % per annum for a period of 5 years. [2]

(d) A bank offers loans to farmers at a compound interest rate of 4,5% per annum.

Calculate the total amount to be paid by a farmer who borrows \$ 45 000 for a period of 2 years. [3]

4 **(a)** It is given that $\begin{pmatrix} 3 & a \\ 4 & 2 \end{pmatrix} + 2\begin{pmatrix} -2 & 4 \\ b & -5 \end{pmatrix} = \begin{pmatrix} -1 & 15 \\ 10 & c \end{pmatrix}$.

Find,

(i) a, [1]

(ii) b, [1]

(iii) c. [1]

(b) Matrix $M = \begin{pmatrix} 6 & 3 \\ 1 & 3 \end{pmatrix}$ and matrix $N = \begin{pmatrix} 7 \\ -3 \end{pmatrix}$.

Find,

(i) MN, [2]

(ii) M^{-1}, the inverse of matrix M, [3]

(iii) Hence or otherwise find matrix P for which $MP = \begin{pmatrix} 3 & -6 \\ -4 & -1 \end{pmatrix}$. [4]

5 **Three villages P, Q and R are on level ground.**
Village Q is $1,4\ km$ **from village P on a bearing of** $N\ 45°W$.
Village R is $2\ km$ **from village P on a bearing of** $N\ 60°E$.

(a) **Using ruler and compasses only and a scale of** $1cm$ **to represent** $200\ m$, **construct on a single diagram, the**

(i) positions of Q and R relative to P, [5]

(ii) locus of points equidistant from P and Q, [2]

(iii) locus of points equidistant from P and R. [2]

(b) A borehole B is equidistant from all three villages, P, Q and R.
Mark and label clearly the point B. [1]

(c) Calculate the actual distance of village R from the borehole in km. [2]

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Section B [48 Marks]

Answer any **four** questions from this section.

Each question carries 12 marks.

6 **(a)** Simplify $\frac{3}{4} \times \left(2\frac{1}{2} - \frac{4}{5}\right)$.

Give the answer as a fraction to its lowest terms. [3]

(b) Three pieces of wire of lengths 36cm, 90cm and 108cm are cut into equal pieces. Find the greatest length of wire possible if no wire is left over. [3]

(c) Sarah and Nobuhle shared a packet of sweets in the ratio 3 : 5 respectively. Sarah received 60 sweets.

(i) Express Nobuhle 's share as a percentage of the total number of sweets in the packet. [2]

(ii) Calculate the number of sweets Nobuhle would get if her share is reduced in the ratio 2 : 3. [2]

(d) Evaluate $503_6 - 231_5$ giving the answer in base 6. [2]

7 **(a)** P varies directly with r and inversely with $(q^2 - 3)$.
$P = 10, \ when \ q = 5 \ and \ r = 11.$
Find the

(i) value of k, the constant of variation [2]

(ii) two possible values of q when $r = 9, 2 \ and \ p = 4$. [4]

(b) The price B, of a jacket partly varies as the cost C, of its material and partly as the time, T, the time taken to make it. If it takes 4 hours to make the jacket and with cost of material $\$ 225$, its price is $\$ 2 225$, when the cost of material is $\$ 450$ and it takes 7 hours to make a jacket costing \$8900.

(i) Express B in terms of C, T and two constants m and n, [1]

(ii) Find the value of m and the value of n, [3]

(iii) Find the price of a jacket that is made in 5 hours and the cost of material of \$350. [2]

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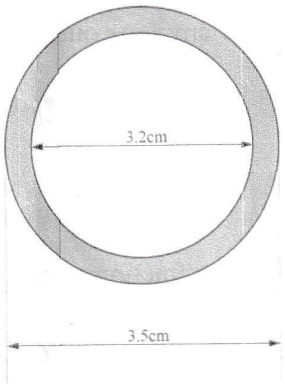
8 **(a)**



The diagram shows the cross section of a hollow cylindrical metal pipe of length 60 cm. The internal and external diameters of the pipe are 3,2 cm and 3,5 cm respectively.

In this part of question take π to be 3,142 .

Calculate the,

(i) circumference of the inner circle, [2]

(ii) internal curved surface area, [2]

(iii) area of the shaded region of the cross section, [4]

(iv) volume of the metal in the pipe. [2]

(b) A cube of solid metal has a mass of 2,42 kg and a volume of 125,3 cm ^{3}.

Calculate the density of metal.

Give the answer in g/cm^3. [2]

9 **Answer the whole of this question on a sheet of graph paper.**

The following is an incomplete table of values for the function $y = \frac{12}{x}$

x	1	$1\frac{1}{2}$	2	3	4	5	6
y	12	m	6	4	3	n	2

(a) Find the value of m and the value of n, [2]

(b) Use a scale of: 2cm to represent 1 unit on the $x - axis$

2 cm to represent 2 units on the $y - axis$,

to draw the graph of $y = \frac{12}{x}$ for $1 \leq x \leq 6$. [3]

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(c) Use the graph to find the,

(i) gradient of the curve at $x = 3$, [2]

(ii) area of the region bounded by the curve, the x - axis and the lines $x = 2$ *and* $x = 4$. [2]

(d) **(i)** On the same axes draw the graph of $y = 3x + 1$. [1]

(ii) Hence use the graph to solve the equation $\frac{12}{x} = 3x + 1$. [2]

10 The heights (h cm) of 30 learners in a class are shown in the table below.

HEIGHT (hcm)	$150 < h \leq 160$	$160 < h \leq 170$	$170 < h \leq 190$	$190 < h \leq 200$
NUMBER OF LEARNERS	6	9	13	2
FREQUENCY DENSITY	0,6	0,9	w 0,4	0,2

(a) Find the value of w. [1]

(b) State the modal class. [1]

(c) Calculate an estimate for the mean height of the learners in the class. [3]

(d) **Use a graph paper to answer this part of the question.**

Using a scale of 2cm to represent 10 units on the horizontal axis and 2cm to represent 0,2 units on the vertical axis, draw a histogram for the data. [4]

(e) If the data is represented on a pie chart, calculate the angle of sector of the class $160 < h \leq 170$. [2]

(f) Find the probability that a learner chosen at random has a height in the range $170 < h \leq 190$. [1]

11 **Answer this question on a sheet of graph paper. Use a scale of 2 cm to represent 1 unit on the x-axis for** $-4 \leq x \leq 5$ **and 2 cm to represent 2 units on the** $y - axis$ **for** $-10 \leq y \leq 12$.

(a) Triangle PQR has vertices at P(2;2), Q(4;6) and R(2;6).

Draw and label triangle PQR. [1]

(b) **(i)** Triangle $P_1Q_1R_1$ has vertices at $P_1\ (-1;4)$, $Q_1\ (-3;8)$ and $R_1\ (-3;4)$.

Draw and label triangle $P_1Q_1R_1$. [1]

(ii) Describe fully a single transformation that maps triangle PQR onto triangle $P_1Q_1R_1$. [3]

(c) Triangle PQR is mapped onto triangle $P_2Q_2R_2$ by an enlargement centre (2;0) and scale factor $-1\frac{1}{2}$. Draw and label triangle $P_2Q_2R_2$. [2]

(d) The matrix $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ represents a transformation which maps triangle PQR onto triangle $P_3Q_3R_3$.

(i) Draw and label triangle $P_3Q_3R_3$. [2]

(ii) Describe fully the single transformation that maps triangle PQR onto triangle $P_3Q_3R_3$. [3]

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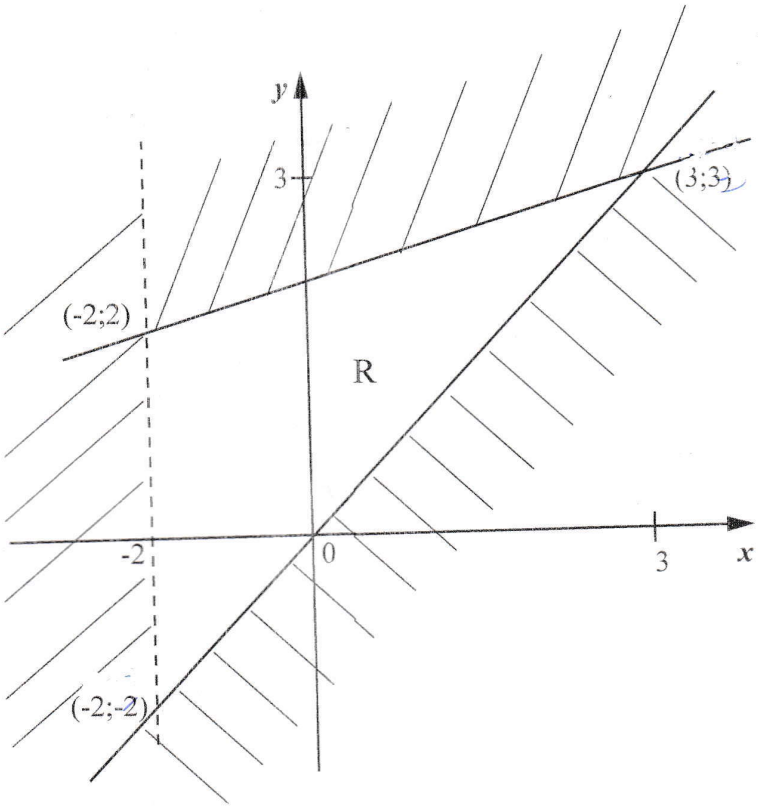
12 **(a)**



The diagram shows region R defined by 3 inequalities.

Find the

(i) three inequalities, [4]

(ii) coordinates of the point that give the maximum value of $2x + y$, [2]

(iii) maximum value of $2x + y$. [2]

(b) A conference is to be attended by both male and female teachers from each school in the district. The number of male teachers should not exceed 9 and at least 5 female teachers from each school should attend. However not more than 15 teachers from each school can attend the conference.

Let x be the number of male teachers and y the number of female teachers from each school who are supposed to attend the conference.

Write down the 3 inequalities that satisfy the above restrictions. [4]